

A negative literature answer to Problem 1 of arXiv:1406.3815

Literature-status note for `fa.banach.001`

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Status. `literature_already_answered` (Problem 1 only).

Original question

In Section 4, Problem 1, page 18 of the arXiv PDF, Nasseri [1] asks:

Is it true that an operator $T : C(K) \rightarrow C(K)$ with dense range, where $C(K)$ has the Grothendieck property, is actually surjective?

Answer

Johnson, Nasseri, Schechtman, and Tkocz [2] answer this negatively. Corollary 1 on page 5 of arXiv:1408.1443 states that there is an injective, dense-range, nonsurjective operator on ℓ_∞ . Since

$$\ell_\infty = C(\beta\mathbb{N})$$

and ℓ_∞ has the Grothendieck property, their operator satisfies every hypothesis of Problem 1 and violates its proposed conclusion.

The construction starts with an injective, dense-range, nonsurjective Tauberian operator S on $L_1(0, 1)$. Tauberianity makes S^{**} injective, so S^* has dense proper range; density of the range of S makes S^* injective. The classical isomorphism $L_1(0, 1)^* \cong \ell_\infty$ transfers S^* to the required operator on ℓ_∞ . This paragraph is only an identification summary; the full construction is in the supporting paper.

The answering paper cites arXiv:1406.3815 and notes, using its spectral theorem, that zero is an interior point of the spectrum of any dense-range, nonsurjective operator on ℓ_∞ .

Scope and search status

This is an exact answer only to Problem 1. Problems 2 (the converse Grothendieck/Dunford–Pettis characterization) and 3 (the H^∞ analogue of the commuting-product corollary) are not answered by arXiv:1408.1443 and are not claimed solved here.

The identification was checked against the original PDFs: Problem 1 occurs on original PDF page 18, and the counterexample is Corollary 1 on supporting PDF page 5. Exact-title, dense-range, Grothendieck, and H^∞ searches through 11 August 2026 found the direct answering paper; no new mathematical result is claimed by this packet.

References

- [1] A. B. Nasser, *The Spectrum of Operators on $C(K)$ with the Grothendieck Property and Characterization of J -Class Operators which are Adjoints*, arXiv:1406.3815 (2014), <https://arxiv.org/abs/1406.3815>.
- [2] W. B. Johnson, A. B. Nasser, G. Schechtman, and T. Tkocz, *Injective Tauberian Operators on L_1 and Operators with Dense Range on ℓ_∞* , Canad. Math. Bull. **58** (2015), 276–280; arXiv:1408.1443, <https://arxiv.org/abs/1408.1443>.